

🌟 Graph Theory & Combinatorics Solutions: Part 2

🌟 Q1 (i) Find the adjacency matrix of the graph given below. Find out the number of paths of length 2 from one vertex to others.

🌟 **Answer:**

♦ **Step 1: Identify Vertices** From Figure 1, the graph has vertices: \$A, B, C, D, E\$

🔍 **Step 2: Identify Edges** From the graph:

- $A \rightarrow B$
- $A \rightarrow C$
- $A \rightarrow D$
- $B \rightarrow D$
- $B \rightarrow E$
- $C \rightarrow D$
- $D \rightarrow E$

Also:

- There is a self-loop at vertex \$D\$.

💡 **Step 3: Construct Adjacency Matrix** Order of vertices: \$A, B, C, D, E\$

Adjacency matrix \$M\$: $M = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

🌟 **Explanation of Entries** Example:

- $M_{12} = 1$ means there is an edge between \$A\$ and \$B\$.
- $M_{45} = 1$ means there is an edge between \$D\$ and \$E\$.
- $M_{44} = 1$ because of the self-loop at \$D\$.

🔥 **Step 4: Paths of Length 2** Important theorem: 🙌 The number of paths of length 2 is obtained from computing \$M^2\$, where \$M\$ is the adjacency matrix.

💡 **Compute \$M^2\$** $M^2 = \begin{bmatrix} 3 & 1 & 1 & 3 & 2 \\ 1 & 3 & 2 & 3 & 1 \\ 1 & 2 & 2 & 2 & 1 \\ 3 & 3 & 2 & 5 & 2 \\ 2 & 1 & 1 & 2 & 2 \end{bmatrix}$

🌈 **Interpretation** Example:

- Entry \$(1,4) = 3\$ means: 🙌 There are **3** different paths of length 2 from \$A\$ to \$D\$.
- Entry \$(2,3) = 2\$ means: 🙌 There are **2** paths of length 2 from \$B\$ to \$C\$.

✅ **Final Answer Adjacency Matrix:** $M = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

Number of paths of length 2: This is formally obtained from the matrix \$M^2\$.

☀️ Q1 (ii) Find out the maximum and minimum number of edges in a 3-component graph having 10 vertices.

🌟 **Answer:**

◆ **Given:**

- Number of vertices: $n = 10$
- Number of components: $k = 3$

☀️ **Minimum Number of Edges** For minimum edges, every component should be a tree. 🌲 We know that for a graph with k components, the minimum number of edges is: $|E|_{\min} = n - k$

Substitute the values: $10 - 3 = 7$

✅ Minimum Edges: ☀️ **7**

☀️ **Maximum Number of Edges** To maximize the edges:

- Make one component as large a complete graph as possible.
- Keep the remaining components as isolated single vertices.

So, the largest component has: $10 - (3 - 1) = 8$ vertices.

The maximum number of edges in a complete graph is given by: $\frac{n(n-1)}{2}$

Hence: $\frac{8(7)}{2} = 28$

✅ Maximum Edges: ☀️ **28**

🎯 **Final Answer:**

Quantity	Answer
Minimum edges	7
Maximum edges	28

☀️ Q2 (i) Does a Hamiltonian path or circuit exist on the graph below (Figure 1)? Justify it.

🌟 **Answer:**

◆ **Hamiltonian Path** A Hamiltonian path visits every vertex exactly once.

◆ **Hamiltonian Circuit** A Hamiltonian circuit starts and ends at the same vertex while visiting every vertex exactly once.

🔍 **Observation** Vertices: A, B, C, D, E Notice that vertex E has a degree of 1.

✗ **Hamiltonian Circuit** A Hamiltonian circuit requires entering and leaving every vertex. But E has only one edge. So, a circuit is impossible. ✗

✅ **Hamiltonian Path** A possible path exists: $B \rightarrow D \rightarrow A \rightarrow C \rightarrow E$

All vertices are visited exactly once. 

Final Conclusion:

Type	Exists?
Hamiltonian Path	 Yes
Hamiltonian Circuit	 No

Reason: The presence of the pendant vertex E makes a circuit impossible.

 Q2 (ii) Prove with a suitable example: "Every tree has either one or two centers".

Answer:

- ◆ **Center of a Tree** The center of a tree is the vertex having the minimum eccentricity.
- ◆ **Eccentricity** The eccentricity of a vertex is the maximum distance from that vertex to any other vertex in the tree.

 **Theorem** Every tree has either one center OR two adjacent centers.

Example 1: One Center

A – B – C – D – E

Distances:

Vertex	Maximum Distance (Eccentricity)
A	4
B	3
C	2
D	3
E	4

The minimum eccentricity is at vertex C . Hence, this tree has exactly one center.

Example 2: Two Centers

A – B – C – D

Eccentricities:

Vertex	Max Distance
A	3
B	2
C	2
D	3

The minimum eccentricity is at vertices \$B\$ and \$C\$. Hence, there are two adjacent centers.



Final Conclusion: Every tree always contains exactly one center OR two adjacent centers.



Q2 (iii) Using Prim's algorithm find the minimal spanning tree of the graph given below (Figure 2).



Answer:

◆ **Prim's Algorithm** Prim's Algorithm:

- starts from any vertex.
- repeatedly selects the smallest available edge.
- strictly avoids forming cycles.



Smallest Edge Weights From the figure:

Edge	Weight
E-G	1
F-C	2
G-H	2
A-B	4
C-H	4
F-G	6
C-D	7
E-F	7
A-E	8
B-C	8
D-J	9
H-J	10
B-E	11
D-H	14

 **Prim's Construction** Choose edges in order:

1. $\$E-G = 1\$$
2. $\$G-H = 2\$$
3. $\$F-C = 2\$$
4. $\$A-B = 4\$$
5. $\$C-H = 4\$$
6. $\$C-D = 7\$$
7. $\$A-E = 8\$$
8. $\$D-J = 9\$$

 **Total Weight:** $\$1 + 2 + 2 + 4 + 4 + 7 + 8 + 9 = 37\$$

 **Final Answer:** The Minimum spanning tree weight is:  **37**

 Q3 (i) Does a cut point and cut edges exist on the graph given below (Figure 3)? If yes then mention them.

 **Answer:**

◆ **Cut Vertex (Cut Point)** A cut vertex is a vertex whose removal disconnects the graph.

 **Observation** In Figure 3:

- Removing vertex $\$d\$$ disconnects the left and right portions.
- Similarly, removing vertex $\$f\$$ disconnects vertex $\$h\$$.

Hence, cut vertices exist.  **Cut Vertices:** $\$d, f\$$

◆ **Cut Edge (Bridge)** A cut edge is an edge whose removal disconnects the graph. The edges $\$f-h\$$ and $\$f-g\$$ act as bridges.

 **Cut Edges:** $\$(f, h), (f, g)\$$

 **Final Answer:**

Type	Elements
Cut vertices	$\$d, f\$$
Cut edges	$\$(f, h), (f, g)\$$

 Q3 (ii) Find the union of the two graphs $\$G_1\$$ and $\$G_2\$$.

 **Answer:**

◆ **Union of Graphs** Union means to:

- combine all vertices.
- combine all edges.

 **Vertex Set** $V(G_1 \cup G_2) = V(G_1) \cup V(G_2)$ It contains: a, b, c, d, e, f

 **Edge Set** Contains all edges from G_1 and G_2 combined together.

 **Final Answer:** The union graph consists of all vertices of both graphs and all edges of both graphs combined.

 Q3 (iii) Justify this using a suitable example: "The vertex connectivity of any graph G can never exceed the edge connectivity of G ."

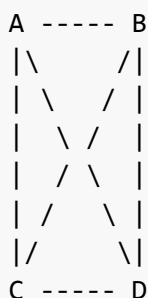
 **Answer:**

◆ **Vertex Connectivity** Vertex connectivity, denoted by $K(G)$, is the minimum number of vertices whose removal disconnects the graph.

◆ **Edge Connectivity** Edge connectivity, denoted by $\lambda(G)$, is the minimum number of edges whose removal disconnects the graph.

 **Important Theorem** For every connected graph: $K(G) \leq \lambda(G)$ Meaning:  Vertex connectivity can never exceed edge connectivity.

 **Suitable Example** Consider a complete graph K_4 :



 **Vertex Connectivity** To disconnect K_4 , at least **3** vertices must be removed. Hence: $K(G) = 3$

 **Edge Connectivity** To disconnect the graph, at least **3** edges must be removed. Hence: $\lambda(G) = 3$

 **Verification** $K(G) = \lambda(G)$ Thus, $K(G) \leq \lambda(G)$ is perfectly satisfied! 

 **Why This Theorem is True:** Removing a vertex automatically removes all incident edges with it. So disconnecting a graph by removing vertices is usually easier (or equally difficult) compared to disconnecting it only by removing edges.

 **Final Conclusion:** For every connected graph, $K(G) \leq \lambda(G)$. Hence proved with a suitable example.

 Q4 (i) "A graph is 2-colorable if it is bipartite and every cycle has an even length." Justify this with a suitable example.

 **Answer:**

♦ **2-Colorable Graph** A graph is called 2-colorable if:

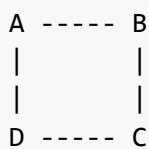
- only two colors are sufficient. 🎨
- adjacent vertices receive different colors.

♦ **Bipartite Graph** A graph is bipartite if:

- vertices can be divided into two disjoint sets.
- edges exist only between the sets.

💡 **Fundamental Theorem** ✅ A graph is bipartite iff every cycle has an even length. Why? Because odd cycles force a same-color conflict ❌, while even cycles allow alternating coloring. ✅

🌈 **Example of Bipartite Graph** Consider a square cycle C_4 :



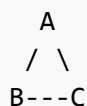
Cycle length: **4**, which is EVEN ✅

🎨 **2-Coloring** Assign colors:

Vertex	Color
A	●
B	●
C	●
D	●

No adjacent vertices have the same color. ✅ Thus, the graph is bipartite and 2-colorable.

❌ **Example of Non-Bipartite Graph** A Triangle:



Cycle length: **3**, which is odd ❌ Using only 2 colors becomes logically impossible here.

🎯 **Final Conclusion:** A graph is 2-colorable exactly when it is bipartite and all cycles are even.

☀️ Q4 (ii) If a graph has 100 vertices and 300 edges, can it be planar?

🌟 **Answer:**

♦ **Planar Graph Condition** For any simple connected planar graph: $e \leq 3v - 6$ Where:

- e = number of edges
- v = number of vertices

🔍 Given

- $v = 100$
- $e = 300$

🔥 **Apply Formula** $3(100) - 6 = 300 - 6 = 294$

For this to be a planar graph, we must have: $e \leq 294$

But we are given $e = 300$, which is strictly greater than **294**. ❌

❌ **Therefore:** The graph cannot be planar.

✅ **Final Answer:** A graph with **100** vertices and **300** edges cannot be planar because it violates the fundamental planar graph condition.

☀️ Q4 (iii) Find the chromatic number of the graph given below (Figure 5).

🌟 **Answer:**

♦ **Chromatic Number** The chromatic number $\chi(G)$ is the minimum number of colors needed to color a graph such that adjacent vertices receive different colors.

🔍 **Observation from Figure 5** The graph shown is a star-like graph containing several intersecting edges and odd cycles. Hence, at least **3** colors are required.

🎨 **Proper Coloring** Use: ● Red, ● Blue, ● Green. Assign colors carefully so adjacent vertices differ. A valid coloring can be obtained using exactly 3 colors.

❌ **Why 2 Colors are Not Enough:** The graph contains odd cycles. Odd cycles are inherently not bipartite. Therefore, the graph is not 2-colorable.

✅ **Final Answer:** $\chi(G) = 3$ The chromatic number is: ☀️ **3**

☀️ Q4 (iv) Find out how many vertices can be matched using maximum matching in the bipartite graph algorithm of Figure 6. Can we call it perfect matching? Justify this.

🌟 **Answer:**

♦ **Matching** A matching is a set of edges where no two edges share a common vertex.

♦ **Maximum Matching** Maximum matching means finding the largest possible matching.

🔍 **Observation from Figure 6**

- Top vertices: \$a, b, c, d, e\$
- Bottom vertices: \$v, w, x, y, z\$

Possible matching:

Top	Bottom
a	w
c	v
d	y
e	z

Total matching edges: **4**

☀️ **Number of Matched Vertices** Each matching edge naturally matches 2 vertices. Thus: $2 \times 4 = 8$

✖️ **Is it Perfect Matching?** Perfect matching means every single vertex must be matched.

- Total vertices: **10**
- Matched vertices: **8**

Two vertices remain unmatched. ✖️ Hence, it is not a perfect matching.

✅ **Final Answer:**

Quantity	Answer
Maximum matched vertices	8
Perfect matching?	✖️ No

Reason: All vertices are not matched.

☀️ Q5 (i) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three hearts are selected?

✨ **Answer:**

♦ **Use Pigeonhole Principle** 🐦 A standard deck contains:

- **13** hearts ❤️
- **39** non-hearts

💡 **Worst Case Scenario** To systematically avoid getting 3 hearts for as long as possible:

- First, select all **39** non-hearts.
- Then, select exactly **2** hearts.

Total cards selected without guaranteeing 3 hearts: $39 + 2 = 41$

Therefore, selecting just one more card guarantees you will draw 3 hearts.

✓ **Final Answer:** $41 + 1 = 42$ 🌟 **42 cards** must be selected.

🌟 Q5 (ii) How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other?

🌟 **Answer:**

♦ **Step 1: Arrange Men** Arrange 8 men: $8!$ ways.

♦ **Step 2: Create Gaps** After arranging the men, the gaps available are: $_ M _ M _ M _ M _ M _ M _ M _ M _$

Number of available gaps: **9**

♦ **Step 3: Choose Positions for Women** Choose 5 gaps from the 9 available: $\binom{9}{5}$

♦ **Step 4: Arrange Women** Arrange the 5 women in those chosen gaps: $5!$

🔥 **Total Arrangements** $8! \times \binom{9}{5} \times 5! = 40320 \times 126 \times 120 = 609638400$

✓ **Final Answer:** 🌟 **609,638,400 ways**

🌟 Q5 (iii) Suppose a club has 25 members. How many ways are there to choose a president, vice-president, secretary, and treasurer where no person holds more than one office?

🌟 **Answer:**

♦ **Important Observation** The different posts are:

- President
- Vice-President
- Secretary
- Treasurer

Because the positions are completely distinct, order matters! ✓ Hence, permutations are used.

💡 **Calculation:**

- President: **25** choices
- Vice-President: **24** choices
- Secretary: **23** choices
- Treasurer: **22** choices

🔥 **Total Ways:** $25 \times 24 \times 23 \times 22 = 303600$

✓ **Final Answer:** 🌟 **303,600 ways**